Instagram Engagement Analysis

Task 3 | Social Media Analytics | Google Colab

7	8 AM	Thursday	#smile
CSV Tables	Best Hour	Best Day	Top Tag

Overview: Multi-table Instagram dataset (users, photos, likes, comments, tags, photo_tags, follows) was merged and analysed to identify engagement drivers for **Alfido Tech.** Feature engineering created *engagement* (likes + comments), *followers count*, and *engagement rate*. Time features (hour, day) were extracted from post timestamps. Key findings: **8 AM** drives peak engagement; **Thursday** is the best posting day. Photo type engagement is similar across photo, video, and carousel. Top hashtags are **#smile, #beach, #party, #fun.** Likes & comments show strong correlation with overall engagement (r = 0.7 & 0.66).

Analysis Charts

```
plt.show()
```

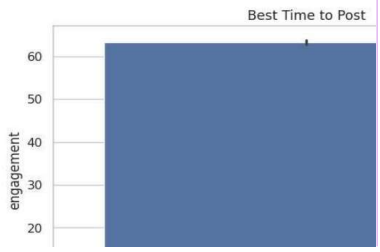

Fig 1 — Best Time to Post

```
plt.show()
```

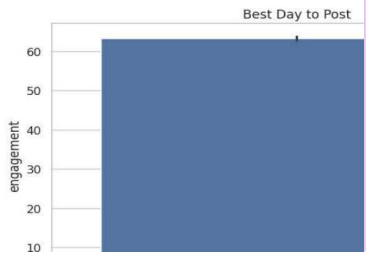

Fig 2 — Best Day to Post

```
plt.title("Content Type vs Engagement")
plt.show()
```

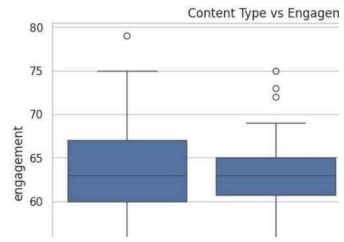

Fig 3 — Content Type vs Engagement

```
tag_counts.plot(kind='bar')
plt.title("Top Hashtags")
plt.show()
```

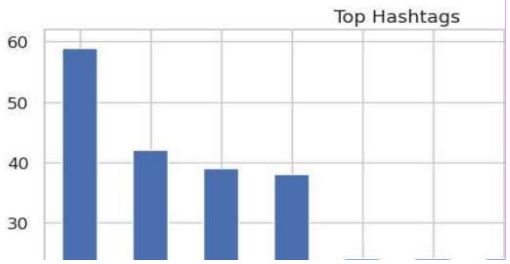

Fig 4 — Top 10 Hashtags by Usage

```
plt.axis('off')
plt.title("Popular Hashtags")
plt.show()
```

Fig 5 — Popular Hashtags Word Cloud

Key Insights

- **8 AM** is peak engagement hour
- **Thursday** drives highest engagement
- Photo, video & carousel perform similarly
- **#smile, #beach, #party** are top tags
- Likes–engagement correlation: **r = 0.70**
- Post **3–5×/week**, avoid Sunday

Content Calendar

Day	Time	Type
Mon	3–5 PM	Educational
Tue	1–5 PM	Reels
Wed	12–4 PM	Carousel
Thu	2–6 PM	Reels/Trends
Fri	3–5 PM	Light Content
Sat	Optional	Stories
Sun	Avoid	—

Notebook & Tools

- **task-3_collab.ipynb** (Colab)
- 7 CSVs: users, photos, likes, comments, tags, photo_tags, follows
- pandas · seaborn · matplotlib · wordcloud
- Merge → Feature Eng. → Analysis → Viz

5 Strategies: Peak Hours · Content Types · Hashtags · Consistency · Interaction